P0AA4-00、 P0AA5-00、 P0AE0-00

Fault diagnosis code
P0AA4-00：Main and negative relay adhesion failure
P0AA5-00：Main negative relay normally open fault
P0AE0-00：Main negative relay drive failure
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• BMS fault • Main negative relay fault • BMS main negative relay circuit fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0AA4-00
Main and negative relay adhesion (more than 95% of front-end voltage for 200ms)
P0AA5-00
The main negative relay is normally open (after the main negative relay is closed and the total voltage 1 is greater than 200V, the difference between the back-end voltage of the main negative relay and the total voltage 1 (battery voltage) is greater than 100V for 2s)
P0AE0-00
Main negative relay drive fault ((It cannot be distinguished from a feedback signal shared by the slave power supply driver chip, and alarms at the same time))
To set the effect of a fault code condition

Troubleshooting steps		
Step 1	General inspection.	
		1 Check battery capacity. 2 Check the harness plug of BMS for damage, poor contact, aging and looseness. 3 Check the harness plug of battery pack high voltage for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 Is there any other fault code except for P0AA4-00/P0AA5-00/P0AE0-00?
		YES → Refer to:DTC Summary list(BMS).
NO ↓		
Step 3	Check the connecting cable of BMS.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of BMS. 5 Check whether the BMS connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 4	Check the main negative relay circuit.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the BMS end cover. 4 Check whether the main negative relay circuit exist the situation of open or short circuit. 5 Check whether the main negative relay circuit exist the situation of shorting to battery. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check main negative relay signal checking module.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the BMS end cover. 4 Check whether the main negative relay signal checking module is damaged. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Check the main negative relay(components).
	1 Check the harness plug of BMS internal main negative relay for damage, poor contact, aging and looseness. 2 Check whether the BMS internal main negative relay works. 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Perform initialization.	
		1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The BMS module was initialized with the diagnostic instrument. 5 Can the BMS be initialized and is the BMS restored?
		NO → Repair or change the BMS module.
YES ↓		
Step 8	Test whether BMS is norm.	
		1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1001 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
		YES → Replace the BMS.
NO ↓		
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.		
Next Step ↓		
Diagnosis end.		